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Research Article

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Effects of *Cordyceps militaris* supplementation on the immune response and upper respiratory infection in healthy adults: a randomized, double-blind, placebo-controlled study

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Abstract

Purpose

Upper respiratory tract infections are major causes of the common cold throughout the world. *Cordyceps militaris* (*C. militaris*) is a well-known functional food for its anti-fatigue and immunomodulating activities. On the other hand, there are no reports on the protective effect against upper respiratory tract infections (URI). This study was a 12 week randomized, double-blind, and placebo-controlled trial in healthy volunteers.

Methods

A total of 100 subjects 20 ~ 70 years of age with a history of at least two colds in the year were enrolled in the study. The participants were required to record any adverse events and rate any cold-related incidents in a diary during the investigation period. The efficacy end point was the symptoms and incidence of URI, and changes in cytokines, IgA and natural killer (NK) cell activity.

Results

The *Cordyceps militaris* group over 12 weeks showed no significant impact on the incidence and symptomatology of URI compared to the placebo group. On the other hand, the experimental group showed significantly higher NK cell activity ($p = 0.047$) and IgA level ($p = 0.035$) compared to the placebo group. The NK-cell activity and IgA level were increased significantly by *Cordyceps militaris* over 12 weeks.

Conclusion

The results suggest the possible beneficial immunomodulating effects, but the protective effects on URI could not be demonstrated under these conditions. Additional research will be needed to determine the efficacy and mechanisms of *Cordyceps militaris* function.

Keywords: Cordyceps militaris, immune, upper respiratory tract infection, NK-cell, IgA

Figures and Tables

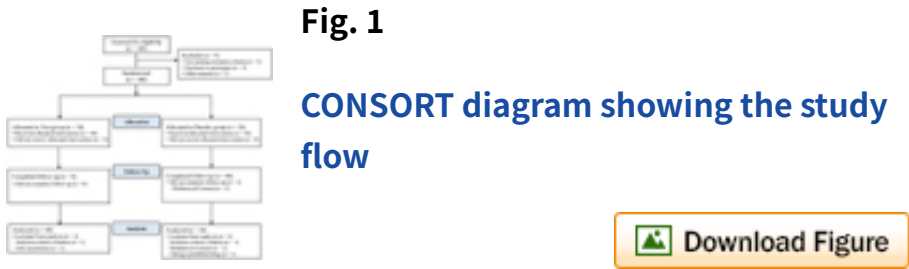


Table 1

General characteristics of subjects

Characteristic	Intervention (n=50)	Control (n=50)
Age (mean ± SD)	23.5 ± 2.1	23.5 ± 2.1
Gender (n)		
Male	25	25
Female	25	25
Height (mean ± SD)	170.5 ± 5.2	170.5 ± 5.2
Weight (mean ± SD)	65.5 ± 10.1	65.5 ± 10.1
Body mass index (mean ± SD)	22.5 ± 2.1	22.5 ± 2.1

Values are presented as mean ± SD or number (%).

1) Analyzed by Student's t test or Chi-square test or Fisher's exact test



Table 2

URI incidence rate of the study participants during the 12

weeks intervention period

Incident		Experimental group (n = 48)	Placebo group (n = 48)	p-value ¹
Wheezing (20-30 breaths/min)	Yes	28 (58.3)	24 (50.0)	0.422
	No	19 (39.6)	24 (50.0)	
Stridor	Yes	17 (35.4)	20 (41.7)	0.424
	No	31 (64.6)	28 (58.3)	
Heart congestion	Yes	15 (31.3)	15 (31.3)	0.988
	No	33 (68.7)	33 (68.7)	
Low blood	Yes	16 (33.3)	14 (29.2)	0.871
	No	32 (66.7)	34 (70.8)	
Cough	Yes	19 (39.6)	14 (29.2)	0.302
	No	29 (60.4)	34 (70.8)	
Sputum	Yes	12 (25.0)	11 (22.9)	0.908
	No	36 (75.0)	37 (77.1)	
Diarrhea	Yes	0 (0.00)	0 (0.00)	-
	No	48 (100.0)	48 (100.0)	
Constipation	Yes	0 (0.0)	0 (0.0)	0.242 ²
	No	48 (100.0)	48 (100.0)	
Headache	Yes	0 (0.00)	0 (0.00)	-
	No	48 (100.0)	48 (100.0)	
Vomiting	Yes	0 (0.00)	1 (2.1)	0.489 ²
	No	48 (100.0)	47 (97.9)	
Headache	Yes	14 (29.2)	5 (10.4)	0.047
	No	34 (70.8)	43 (89.6)	
Malaise	Yes	6 (12.5)	6 (12.5)	0.822
	No	42 (87.5)	42 (87.5)	
Headache	Yes	12 (25.0)	12 (25.0)	0.899
	No	36 (75.0)	36 (75.0)	
Swelling	Yes	7 (14.6)	4 (8.3)	0.548
	No	41 (85.4)	44 (91.7)	
Diarrhea	Yes	1 (2.1)	0 (0.0)	1.000
	No	47 (97.9)	48 (100.0)	

Values are presented as number (%).

1) \$, p-value by Chi-square test; ¥, p-value by Fisher's exact test

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Table 3

Parameters of each group at baseline (0 week) and after 12 weeks intervention

	Experimental group (n = 48)			Placebo group (n = 48)			
	Baseline	12 weeks	SD	Baseline	12 weeks	SD	p-value ¹
Wheezing (20-30 breaths/min)	28.0 ± 1.5	28.0 ± 1.5	1.5	24.0 ± 1.5	24.0 ± 1.5	1.5	0.422
Stridor	17.0 ± 1.5	17.0 ± 1.5	1.5	20.0 ± 1.5	20.0 ± 1.5	1.5	0.424
Heart congestion	15.0 ± 1.5	15.0 ± 1.5	1.5	15.0 ± 1.5	15.0 ± 1.5	1.5	0.988
Low blood	16.0 ± 1.5	16.0 ± 1.5	1.5	14.0 ± 1.5	14.0 ± 1.5	1.5	0.871
Cough	19.0 ± 1.5	19.0 ± 1.5	1.5	14.0 ± 1.5	14.0 ± 1.5	1.5	0.302
Sputum	12.0 ± 1.5	12.0 ± 1.5	1.5	11.0 ± 1.5	11.0 ± 1.5	1.5	0.908
Diarrhea	0.0 ± 0.0	0.0 ± 0.0	0.0	0.0 ± 0.0	0.0 ± 0.0	0.0	-
Constipation	0.0 ± 0.0	0.0 ± 0.0	0.0	0.0 ± 0.0	0.0 ± 0.0	0.0	0.242 ²
Headache	0.0 ± 0.0	0.0 ± 0.0	0.0	0.0 ± 0.0	0.0 ± 0.0	0.0	-
Vomiting	0.0 ± 0.0	0.0 ± 0.0	0.0	1.0 ± 0.0	1.0 ± 0.0	0.0	0.489 ²
Headache	14.0 ± 1.5	14.0 ± 1.5	1.5	5.0 ± 1.5	5.0 ± 1.5	1.5	0.047
Malaise	6.0 ± 1.5	6.0 ± 1.5	1.5	6.0 ± 1.5	6.0 ± 1.5	1.5	0.822
Headache	12.0 ± 1.5	12.0 ± 1.5	1.5	12.0 ± 1.5	12.0 ± 1.5	1.5	0.899
Swelling	7.0 ± 1.5	7.0 ± 1.5	1.5	4.0 ± 1.5	4.0 ± 1.5	1.5	0.548
Diarrhea	1.0 ± 0.0	1.0 ± 0.0	0.0	0.0 ± 0.0	0.0 ± 0.0	0.0	1.000
	47.0 ± 1.5	47.0 ± 1.5	1.5	48.0 ± 1.5	48.0 ± 1.5	1.5	

Data are presented as the mean $\pm$ SD.

1) Analyzed by paired t-test between baseline and 12 weeks in each group

2) Analyzed by linear mixed-effect model and the p-value

represents the comparison to the placebo group

SBP, Systolic blood pressure; DBP, Diastolic blood pressure; NK cell, Natural killer cells; IFN- γ , Interferon gamma; IL-2, Interleukin 2; IgA, Immunoglobulin A

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Table 4

Safety outcome measures between Experimental group and placebo group at baseline and after 12weeks intervention

	Experimental group (n = 10)		Placebo group (n = 10)		p-value
SBP (mmHg)	110.5 ± 12.5	107.5 ± 11.5	111.5 ± 13.5	109.5 ± 12.5	0.12
DBP (mmHg)	75.5 ± 8.5	74.5 ± 7.5	76.5 ± 9.5	75.5 ± 8.5	0.15
HR (b/min)	72.5 ± 10.5	71.5 ± 9.5	73.5 ± 11.5	72.5 ± 10.5	0.18
ECG (mm)	1.2 ± 0.2	1.1 ± 0.2	1.3 ± 0.3	1.2 ± 0.2	0.20
ECG (ms)	100.5 ± 15.5	99.5 ± 14.5	101.5 ± 16.5	100.5 ± 15.5	0.22
ECG (ms)	110.5 ± 16.5	109.5 ± 15.5	111.5 ± 17.5	110.5 ± 16.5	0.24
ECG (ms)	120.5 ± 17.5	119.5 ± 16.5	121.5 ± 18.5	120.5 ± 17.5	0.26
ECG (ms)	130.5 ± 18.5	129.5 ± 17.5	131.5 ± 19.5	130.5 ± 18.5	0.28
ECG (ms)	140.5 ± 19.5	139.5 ± 18.5	141.5 ± 20.5	140.5 ± 19.5	0.30
ECG (ms)	150.5 ± 20.5	149.5 ± 19.5	151.5 ± 21.5	150.5 ± 20.5	0.32
ECG (ms)	160.5 ± 21.5	159.5 ± 20.5	161.5 ± 22.5	160.5 ± 21.5	0.34
ECG (ms)	170.5 ± 22.5	169.5 ± 21.5	171.5 ± 23.5	170.5 ± 22.5	0.36
ECG (ms)	180.5 ± 23.5	179.5 ± 22.5	181.5 ± 24.5	180.5 ± 23.5	0.38
ECG (ms)	190.5 ± 24.5	189.5 ± 23.5	191.5 ± 25.5	190.5 ± 24.5	0.40
ECG (ms)	200.5 ± 25.5	199.5 ± 24.5	201.5 ± 26.5	200.5 ± 25.5	0.42
ECG (ms)	210.5 ± 26.5	209.5 ± 25.5	211.5 ± 27.5	210.5 ± 26.5	0.44
ECG (ms)	220.5 ± 27.5	219.5 ± 26.5	221.5 ± 28.5	220.5 ± 27.5	0.46
ECG (ms)	230.5 ± 28.5	229.5 ± 27.5	231.5 ± 29.5	230.5 ± 28.5	0.48
ECG (ms)	240.5 ± 29.5	239.5 ± 28.5	241.5 ± 30.5	240.5 ± 29.5	0.50
ECG (ms)	250.5 ± 30.5	249.5 ± 29.5	251.5 ± 31.5	250.5 ± 30.5	0.52
ECG (ms)	260.5 ± 31.5	259.5 ± 30.5	261.5 ± 32.5	260.5 ± 31.5	0.54
ECG (ms)	270.5 ± 32.5	269.5 ± 31.5	271.5 ± 33.5	270.5 ± 32.5	0.56
ECG (ms)	280.5 ± 33.5	279.5 ± 32.5	281.5 ± 34.5	280.5 ± 33.5	0.58
ECG (ms)	290.5 ± 34.5	289.5 ± 33.5	291.5 ± 35.5	290.5 ± 34.5	0.60
ECG (ms)	300.5 ± 35.5	299.5 ± 34.5	301.5 ± 36.5	300.5 ± 35.5	0.62
ECG (ms)	310.5 ± 36.5	309.5 ± 35.5	311.5 ± 37.5	310.5 ± 36.5	0.64
ECG (ms)	320.5 ± 37.5	319.5 ± 36.5	321.5 ± 38.5	320.5 ± 37.5	0.66
ECG (ms)	330.5 ± 38.5	329.5 ± 37.5	331.5 ± 39.5	330.5 ± 38.5	0.68
ECG (ms)	340.5 ± 39.5	339.5 ± 38.5	341.5 ± 40.5	340.5 ± 39.5	0.70
ECG (ms)	350.5 ± 40.5	349.5 ± 39.5	351.5 ± 41.5	350.5 ± 40.5	0.72
ECG (ms)	360.5 ± 41.5	359.5 ± 40.5	361.5 ± 42.5	360.5 ± 41.5	0.74
ECG (ms)	370.5 ± 42.5	369.5 ± 41.5	371.5 ± 43.5	370.5 ± 42.5	0.76
ECG (ms)	380.5 ± 43.5	379.5 ± 42.5	381.5 ± 44.5	380.5 ± 43.5	0.78
ECG (ms)	390.5 ± 44.5	389.5 ± 43.5	391.5 ± 45.5	390.5 ± 44.5	0.80
ECG (ms)	400.5 ± 45.5	399.5 ± 44.5	401.5 ± 46.5	400.5 ± 45.5	0.82
ECG (ms)	410.5 ± 46.5	409.5 ± 45.5	411.5 ± 47.5	410.5 ± 46.5	0.84
ECG (ms)	420.5 ± 47.5	419.5 ± 46.5	421.5 ± 48.5	420.5 ± 47.5	0.86
ECG (ms)	430.5 ± 48.5	429.5 ± 47.5	431.5 ± 49.5	430.5 ± 48.5	0.88
ECG (ms)	440.5 ± 49.5	439.5 ± 48.5	441.5 ± 50.5	440.5 ± 49.5	0.90
ECG (ms)	450.5 ± 50.5	449.5 ± 49.5	451.5 ± 51.5	450.5 ± 50.5	0.92
ECG (ms)	460.5 ± 51.5	459.5 ± 50.5	461.5 ± 52.5	460.5 ± 51.5	0.94
ECG (ms)	470.5 ± 52.5	469.5 ± 51.5	471.5 ± 53.5	470.5 ± 52.5	0.96
ECG (ms)	480.5 ± 53.5	479.5 ± 52.5	481.5 ± 54.5	480.5 ± 53.5	0.98
ECG (ms)	490.5 ± 54.5	489.5 ± 53.5	491.5 ± 55.5	490.5 ± 54.5	1.00
ECG (ms)	500.5 ± 55.5	499.5 ± 54.5	501.5 ± 56.5	500.5 ± 55.5	1.02
ECG (ms)	510.5 ± 56.5	509.5 ± 55.5	511.5 ± 57.5	510.5 ± 56.5	1.04
ECG (ms)	520.5 ± 57.5	519.5 ± 56.5	521.5 ± 58.5	520.5 ± 57.5	1.06
ECG (ms)	530.5 ± 58.5	529.5 ± 57.5	531.5 ± 59.5	530.5 ± 58.5	1.08
ECG (ms)	540.5 ± 59.5	539.5 ± 58.5	541.5 ± 60.5	540.5 ± 59.5	1.10
ECG (ms)	550.5 ± 60.5	549.5 ± 59.5	551.5 ± 61.5	550.5 ± 60.5	1.12
ECG (ms)	560.5 ± 61.5	559.5 ± 60.5	561.5 ± 62.5	560.5 ± 61.5	1.14
ECG (ms)	570.5 ± 62.5	569.5 ± 61.5	571.5 ± 63.5	570.5 ± 62.5	1.16
ECG (ms)	580.5 ± 63.5	579.5 ± 62.5	581.5 ± 64.5	580.5 ± 63.5	1.18
ECG (ms)	590.5 ± 64.5	589.5 ± 63.5	591.5 ± 65.5	590.5 ± 64.5	1.20
ECG (ms)	600.5 ± 65.5	599.5 ± 64.5	601.5 ± 66.5	600.5 ± 65.5	1.22
ECG (ms)	610.5 ± 66.5	609.5 ± 65.5	611.5 ± 67.5	610.5 ± 66.5	1.24
ECG (ms)	620.5 ± 67.5	619.5 ± 66.5	621.5 ± 68.5	620.5 ± 67.5	1.26
ECG (ms)	630.5 ± 68.5	629.5 ± 67.5	631.5 ± 69.5	630.5 ± 68.5	1.28
ECG (ms)	640.5 ± 69.5	639.5 ± 68.5	641.5 ± 70.5	640.5 ± 69.5	1.30
ECG (ms)	650.5 ± 70.5	649.5 ± 69.5	651.5 ± 71.5	650.5 ± 70.5	1.32
ECG (ms)	660.5 ± 71.5	659.5 ± 70.5	661.5 ± 72.5	660.5 ± 71.5	1.34
ECG (ms)	670.5 ± 72.5	669.5 ± 71.5	671.5 ± 73.5	670.5 ± 72.5	1.36
ECG (ms)	680.5 ± 73.5	679.5 ± 72.5	681.5 ± 74.5	680.5 ± 73.5	1.38
ECG (ms)	690.5 ± 74.5	689.5 ± 73.5	691.5 ± 75.5	690.5 ± 74.5	1.40
ECG (ms)	700.5 ± 75.5	699.5 ± 74.5	701.5 ± 76.5	700.5 ± 75.5	1.42
ECG (ms)	710.5 ± 76.5	709.5 ± 75.5	711.5 ± 77.5	710.5 ± 76.5	1.44
ECG (ms)	720.5 ± 77.5	719.5 ± 76.5	721.5 ± 78.5	720.5 ± 77.5	1.46
ECG (ms)	730.5 ± 78.5	729.5 ± 77.5	731.5 ± 79.5	730.5 ± 78.5	1.48
ECG (ms)	740.5 ± 79.5	739.5 ± 78.5	741.5 ± 80.5	740.5 ± 79.5	1.50
ECG (ms)	750.5 ± 80.5	749.5 ± 79.5	751.5 ± 81.5	750.5 ± 80.5	1.52
ECG (ms)	760.5 ± 81.5	759.5 ± 80.5	761.5 ± 82.5	760.5 ± 81.5	1.54
ECG (ms)	770.5 ± 82.5	769.5 ± 81.5	771.5 ± 83.5	770.5 ± 82.5	1.56
ECG (ms)	780.5 ± 83.5	779.5 ± 82.5	781.5 ± 84.5	780.5 ± 83.5	1.58
ECG (ms)	790.5 ± 84.5	789.5 ± 83.5	791.5 ± 85.5	790.5 ± 84.5	1.60
ECG (ms)	800.5 ± 85.5	799.5 ± 84.5	801.5 ± 86.5	800.5 ± 85.5	1.62
ECG (ms)	810.5 ± 86.5	809.5 ± 85.5	811.5 ± 87.5	810.5 ± 86.5	1.64
ECG (ms)	820.5 ± 87.5	819.5 ± 86.5	821.5 ± 88.5	820.5 ± 87.5	1.66
ECG (ms)	830.5 ± 88.5	829.5 ± 87.5	831.5 ± 89.5	830.5 ± 88.5	1.68
ECG (ms)	840.5 ± 89.5	839.5 ± 88.5	841.5 ± 90.5	840.5 ± 89.5	1.70
ECG (ms)	850.5 ± 90.5	849.5 ± 89.5	851.5 ± 91.5	850.5 ± 90.5	1.72
ECG (ms)	860.5 ± 91.5	859.5 ± 90.5	861.5 ± 92.5	860.5 ± 91.5	1.74
ECG (ms)	870.5 ± 92.5	869.5 ± 91.5	871.5 ± 93.5	870.5 ± 92.5	1.76
ECG (ms)	880.5 ± 93.5	879.5 ± 92.5	881.5 ± 94.5	880.5 ± 93.5	1.78
ECG (ms)	890.5 ± 94.5	889.5 ± 93.5	891.5 ± 95.5	890.5 ± 94.5	1.80
ECG (ms)	900.5 ± 95.5	899.5 ± 94.5	901.5 ± 96.5	900.5 ± 95.5	1.82
ECG (ms)	910.5 ± 96.5	909.5 ± 95.5	911.5 ± 97.5	910.5 ± 96.5	1.84
ECG (ms)	920.5 ± 97.5	919.5 ± 96.5	921.5 ± 98.5	920.5 ± 97.5	1.86
ECG (ms)	930.5 ± 98.5	929.5 ± 97.5	931.5 ± 99.5	930.5 ± 98.5	1.88
ECG (ms)	940.5 ± 99.5	939.5 ± 98.5	941.5 ± 100.5	940.5 ± 99.5	1.90
ECG (ms)	950.5 ± 100.5	949.5 ± 99.5	951.5 ± 101.5	950.5 ± 100.5	1.92
ECG (ms)	960.5 ± 101.5	959.5 ± 100.5	961.5 ± 102.5	960.5 ± 101.5	1.94
ECG (ms)	970.5 ± 102.5	969.5 ± 101.5	971.5 ± 103.5	970.5 ± 102.5	1.96
ECG (ms)	980.5 ± 103.5	979.5 ± 102.5	981.5 ± 104.5	980.5 ± 103.5	1.98
ECG (ms)	990.5 ± 104.5	989.5 ± 103.5	991.5 ± 105.5	990.5 ± 104.5	2.00
ECG (ms)	1000.5 ± 105.5	999.5 ± 104.5	1001.5 ± 106.5	1000.5 ± 105.5	2.02
ECG (ms)	1010.5 ± 106.5	1009.5 ± 105.5	1011.5 ± 107.5	1010.5 ± 106.5	2.04
ECG (ms)	1020.5 ± 107.5	1019.5 ± 106.5	1021.5 ± 108.5	1020.5 ± 107.5	2.06
ECG (ms)	1030.5 ± 108.5	1029.5 ± 107.5	1031.5 ± 109.5	1030.5 ± 108.5	2.08
ECG (ms)	1040.5 ± 109.5	1039.5 ± 108.5	1041.5 ± 110.5	1040.5 ± 109.5	2.10
ECG (ms)	1050.5 ± 110.5	1049.5 ± 109.5	1051.5 ± 111.5	1050.5 ± 110.5	2.12
ECG (ms)	1060.5 ± 111.5	1059.5 ± 110.5	1061.5 ± 112.5	1060.5 ± 111.5	2.14
ECG (ms)	1070.5 ± 112.5	1069.5 ± 111.5	1071.5 ± 113.5	1070.5 ± 112.5	2.16
ECG (ms)	1080.5 ± 113.5	1079.5 ± 112.5	1081.5 ± 114.5	1080.5 ± 113.5	2.18
ECG (ms)	1090.5 ± 114.5	1089.5 ± 113.5	1091.5 ± 115.5	1090.5 ± 114.5	2.20
ECG (ms)	1100.5 ± 115.5	1099.5 ± 114.5	1101.5 ± 116.5	1100.5 ± 115.5	2.22
ECG (ms)	1110.5 ± 116.5	1109.5 ± 115.5	1111.5 ± 117.5	1110.5 ± 116.5	2.24
ECG (ms)	1120.5 ± 117.5	1119.5 ± 116.5	1121.5 ± 118.5	1120.5 ± 117.5	2.26
ECG (ms)	1130.5 ± 118.5	1129.5 ± 117.5	1131.5 ± 119.5	1130.5 ± 118.5	2.28
ECG (ms)	1140.5 ± 119.5	1139.5 ± 118.5	1141.5 ± 120.5	1140.5 ± 119.5	2.30
ECG (ms)	1150.5 ± 120.5	1149.5 ± 119.5	1151.5 ± 121.5	1150.5 ± 120.5	2.32
ECG (ms)	1160.5 ± 121.5	1159.5 ± 120.5	1161.5 ± 122.5	1160.5 ± 121.5	2.34
ECG (ms)	1170.5 ± 122.5	1169.5 ± 121.5	1171.5 ± 123.5	1170.5 ± 122.5	2.36
ECG (ms)	1180.5 ± 123.5	1179.5 ± 122.5	1181.5 ± 124.5	1180.5 ± 123.5	2.38
ECG (ms)	1190.5 ± 124.5	1189.5 ± 123.5	1191.5 ± 125.5	1190.5 ± 124.5	2.40
ECG (ms)	1200.5 ± 125.5	1199.5 ± 124.5	1201.5 ± 126.5	1200.5 ± 125.5	2.42
ECG (ms)	1210.5 ± 126.5	1209.5 ± 125.5	1211.5 ± 127.5	1210.5 ± 126.5	2.44
ECG (ms)	1220.5 ± 127.5	1219.5 ± 126.5	1221.5 ± 128.5	1220.5 ± 127.5	2.46
ECG (ms)	1230.5 ± 128.5	1229.5 ± 127.5	1231.5 ± 129.5	1230.5 ± 128.5	2.48
ECG (ms)	1240.5 ± 129.5	1239.5 ± 128.5	1241.5 ± 130.5	1240.5 ± 129.5	2.50
ECG (ms)	1250.5 ± 130.5	1249.5 ± 129.5	1251.5 ± 131.5	1250.5 ± 130.5	2.52
ECG (ms)	1260.5 ± 131.5	1259.5 ± 130.5	1261.5 ± 132.5	1260.5 ± 131.5	2.54
ECG (ms)	1270.5 ± 132.5	1269.5 ± 131.5	1271.5 ± 133.5	1270.5 ± 132.5	2.56
ECG (ms)	1280.5 ± 133.5	1279.5 ± 132.5	1281.5 ± 134.5	1280.5 ± 133.5	2.58
ECG (ms)	1290.5 ± 134.5	1289.5 ± 133.5	1291.5 ± 135.5	1290.5 ± 134.5	2.60
ECG (ms)	1300.5 ± 135.5	1299.5 ± 134.5	1301.5 ± 136.5	1300.5 ± 135.5	2.62
ECG (ms)	1310.5 ± 136.5	1309.5 ± 135.5	1311.5 ± 137.5	1310.5 ± 136.5	2.64
ECG (ms)	1320.5 ± 137.5	1319.5 ± 136.5	1321.5 ± 138.5	1320.5 ± 137.5	2.66
ECG (ms)	1330.5 ± 138.5	1329.5 ± 137.5	1331.5 ± 139.5	1330.5 ± 138.5	2.68
ECG (ms)	1340.5 ± 139.5	1339.5 ± 138.5	1341.5 ± 140.5	1340.5 ± 139.5	2.70
ECG (ms)	1350.5 ± 140.5	1349.5 ± 139.5	1351.5 ± 141.5	1350.5 ± 140.5	2.72
ECG (ms)	1360.5 ± 141.5	1359.5 ± 140.5	1361.5 ± 142.5	1360.5 ± 141.5	2.74
ECG (ms)	1370.5 ± 142.5	1369.5 ± 141.5	1371.5 ± 143.5	1370.5 ± 142.5	2.76
ECG (ms)	1380.5 ± 143.5	1379.5 ± 142.5	1381.5 ± 144.5	1380.5 ± 143.5	2.78
ECG (ms)	1390.5 ± 144.5	1389.5 ± 143.5	1391.5 ± 145.5	1390.5 ± 144.5	2.80
ECG (ms)	1400.5 ± 145.5	1399.5 ± 144.5	1401.5 ± 146.5	1400.5 ± 145.5	2.82
ECG (ms)	1410.5 ± 146.5	1409.5 ± 145.5	1411.5 ± 147.5	1410.5 ± 146.5	2.84
ECG (ms)	1420.5 ± 147.5	1419.5 ± 146.5	1421.5 ± 148.5	1420.5 ± 147.5	2.86
ECG (ms)	1430.5 ± 148.5	1429.5 ± 147.5	1431.5 ± 149.5	1430.5 ± 148.5	2.88
ECG (ms)	1440.5 ± 149.5	1439.5 ± 148.5	1441.5 ± 150.5	1440.5 ± 149.5	2.90
ECG (ms)	1450.5 ± 150.5				

Table 5

Dietary intake of the study participants during the 12 weeks intervention period

	Baseline	12 weeks	p-value
Energy (kcal)	2050 ± 250	2050 ± 250	0.95
Protein (g)	75 ± 10	75 ± 10	0.95
Carbohydrate (g)	250 ± 30	250 ± 30	0.95
Fat (g)	70 ± 10	70 ± 10	0.95
Fiber (g)	25 ± 5	25 ± 5	0.95
Vitamin A (μg)	1000 ± 100	1000 ± 100	0.95
Vitamin B1 (mg)	1.5 ± 0.2	1.5 ± 0.2	0.95
Vitamin B2 (mg)	1.2 ± 0.1	1.2 ± 0.1	0.95
Vitamin B6 (mg)	1.0 ± 0.1	1.0 ± 0.1	0.95
Vitamin C (mg)	100 ± 20	100 ± 20	0.95
Vitamin E (mg)	10 ± 2	10 ± 2	0.95
Vitamin K (μg)	100 ± 20	100 ± 20	0.95
Calcium (mg)	1000 ± 100	1000 ± 100	0.95
Iron (mg)	10 ± 2	10 ± 2	0.95
Zinc (mg)	10 ± 2	10 ± 2	0.95
Selenium (μg)	100 ± 20	100 ± 20	0.95
Copper (mg)	1.0 ± 0.1	1.0 ± 0.1	0.95
Manganese (mg)	1.0 ± 0.1	1.0 ± 0.1	0.95
Magnesium (mg)	100 ± 20	100 ± 20	0.95
Potassium (mg)	1000 ± 100	1000 ± 100	0.95
Sodium (mg)	1000 ± 100	1000 ± 100	0.95
Phosphorus (mg)	1000 ± 100	1000 ± 100	0.95
Chlorine (mg)	1000 ± 100	1000 ± 100	0.95
Sulfur (mg)	1000 ± 100	1000 ± 100	0.95
Fluorine (mg)	1000 ± 100	1000 ± 100	0.95
Iodine (μg)	100 ± 20	100 ± 20	0.95
Chromium (μg)	100 ± 20	100 ± 20	0.95
Molybdenum (μg)	100 ± 20	100 ± 20	0.95
Cobalt (μg)	100 ± 20	100 ± 20	0.95
Nickel (μg)	100 ± 20	100 ± 20	0.95
Vanadium (μg)	100 ± 20	100 ± 20	0.95
Silicon (mg)	1000 ± 100	1000 ± 100	0.95
Boron (μg)	100 ± 20	100 ± 20	0.95
Aluminum (mg)	1000 ± 100	1000 ± 100	0.95
Barium (μg)	100 ± 20	100 ± 20	0.95
Bismuth (μg)	100 ± 20	100 ± 20	0.95
Bromine (μg)	100 ± 20	100 ± 20	0.95
Cadmium (μg)	100 ± 20	100 ± 20	0.95
Cerium (μg)	100 ± 20	100 ± 20	0.95
Chlorine (mg)	1000 ± 100	1000 ± 100	0.95
Copper (mg)	1.0 ± 0.1	1.0 ± 0.1	0.95
Fluorine (mg)	1000 ± 100	1000 ± 100	0.95
Gold (μg)	100 ± 20	100 ± 20	0.95
Hydrogen (g)	1000 ± 100	1000 ± 100	0.95
Iodine (μg)	100 ± 20	100 ± 20	0.95
Iron (mg)	10 ± 2	10 ± 2	0.95
Lithium (μg)	100 ± 20	100 ± 20	0.95
Magnesium (mg)	100 ± 20	100 ± 20	0.95
Manganese (mg)	1.0 ± 0.1	1.0 ± 0.1	0.95
Molybdenum (μg)	100 ± 20	100 ± 20	0.95
Nickel (μg)	100 ± 20	100 ± 20	0.95
Phosphorus (mg)	1000 ± 100	1000 ± 100	0.95
Potassium (mg)	1000 ± 100	1000 ± 100	0.95
Selenium (μg)	100 ± 20	100 ± 20	0.95
Silver (μg)	100 ± 20	100 ± 20	0.95
Sulfur (mg)	1000 ± 100	1000 ± 100	0.95
Tellurium (μg)	100 ± 20	100 ± 20	0.95
Vanadium (μg)	100 ± 20	100 ± 20	0.95
Zinc (mg)	10 ± 2	10 ± 2	0.95
Zirconium (μg)	100 ± 20	100 ± 20	0.95

Data are presented as the mean ± SD.

- 1) Analyzed by paired t-test between baseline and 12 weeks in each group
- 2) Analyzed by linear mixed-effect model and the p-value represents the comparison to the placebo group



Notes

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